failure1 — Native-IMU reconstruction: method and interpretation

Separate experiment: existing mocap identification and outputs are unchanged.

Identification uses native gyro / specific force with fitted thrust, gyro and accel first-order lags.

Orientation: gyro prediction + acceleration update recur; magnetic factors affect output only.

Only initial orientation / position and initial velocity use mocap. Later mocap is read-only evaluation.

Comparison uses original mocap timestamps inside IMU support; reconstructed attitudes are Slerped.

Native timestamps and firmware first-or-changed magnetic freshness proxy are retained.

Static sensor covariance = trace(unbiased static covariance)/3 times identity; no bias subtraction.

Offline local polynomial derivatives and their overlap covariance are included in the lag fit.

Uncertainty: Bingham-derived right-angle covariance and deterministic local Gaussian propagation.

Shared acceleration/attitude information uses declared conservative covariance bounds.

These algebraic bounds do not guarantee coverage of systematic acceleration or magnetic bias.

Magnetic output can be worse on these data; no mocap-based gating, tuning or recursive feedback.

Model reconstruction adds the native-IMU residual wrench time series to the physical dynamics.

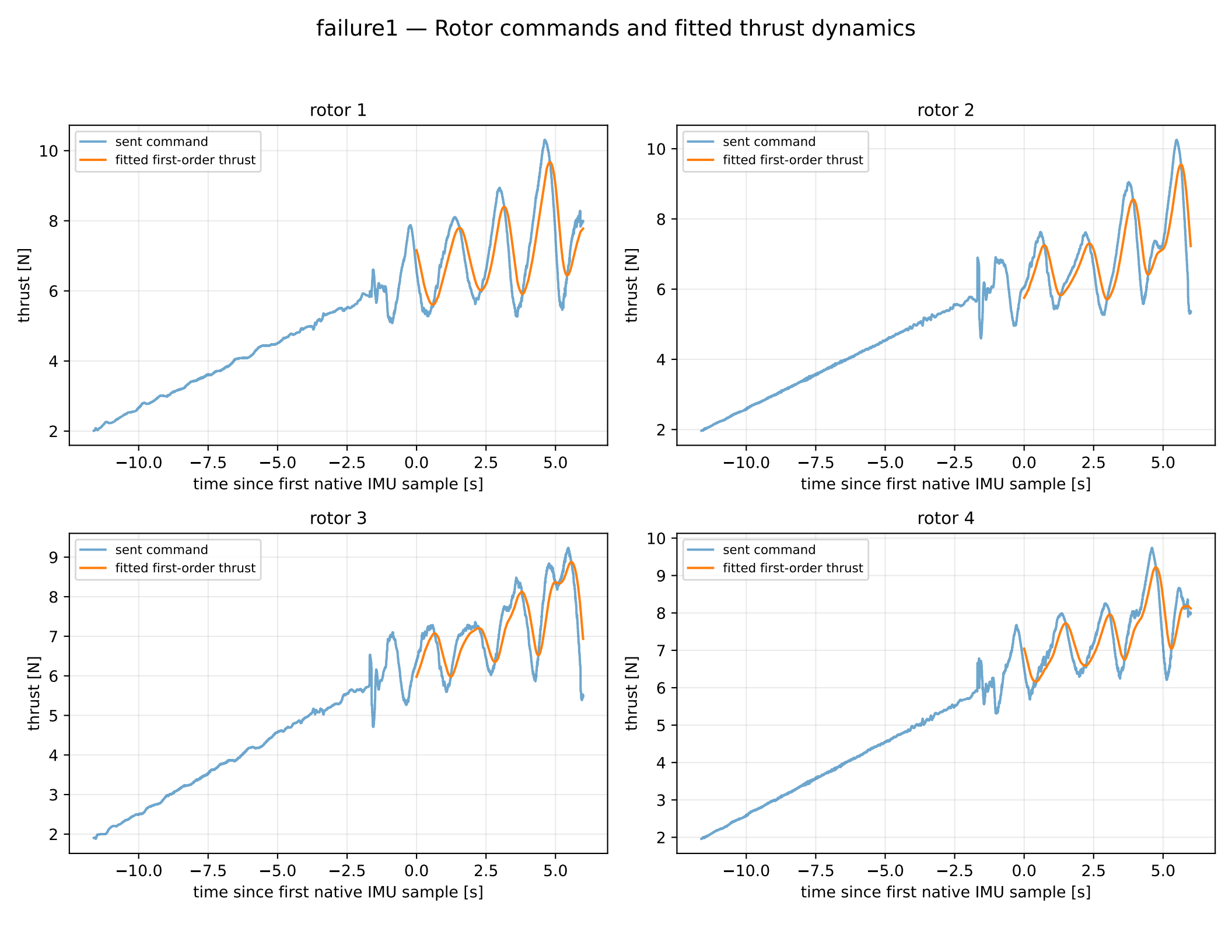
No Monte Carlo, artificial display variance, eigenvalue clipping or uncertainty-based sample selection.

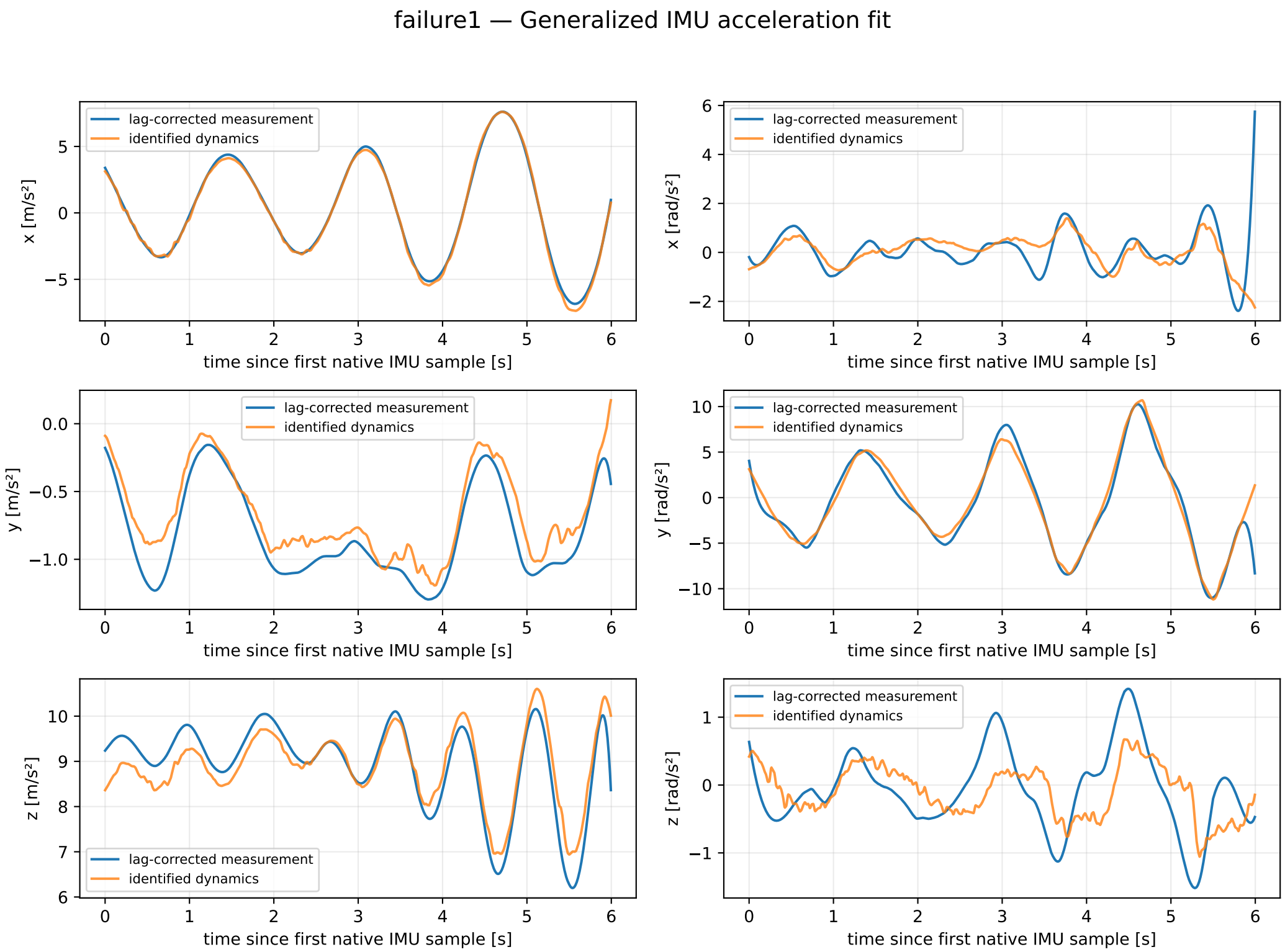
Mass / inertia / thrust scale use a common-scale gauge; chip lever arm is fixed, not calibrated.

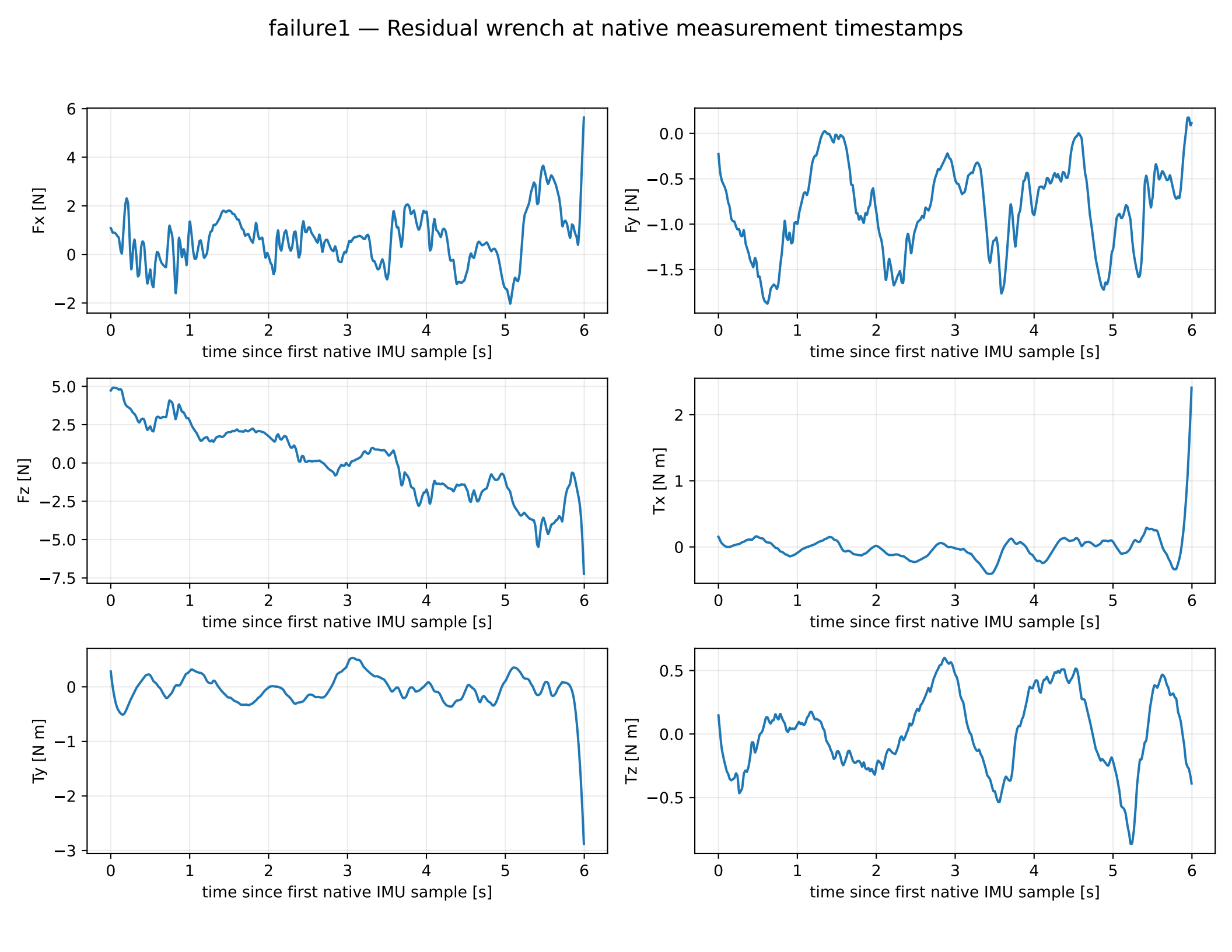
All 17 actual SI estimates and the complete joint covariance are appended at the end.

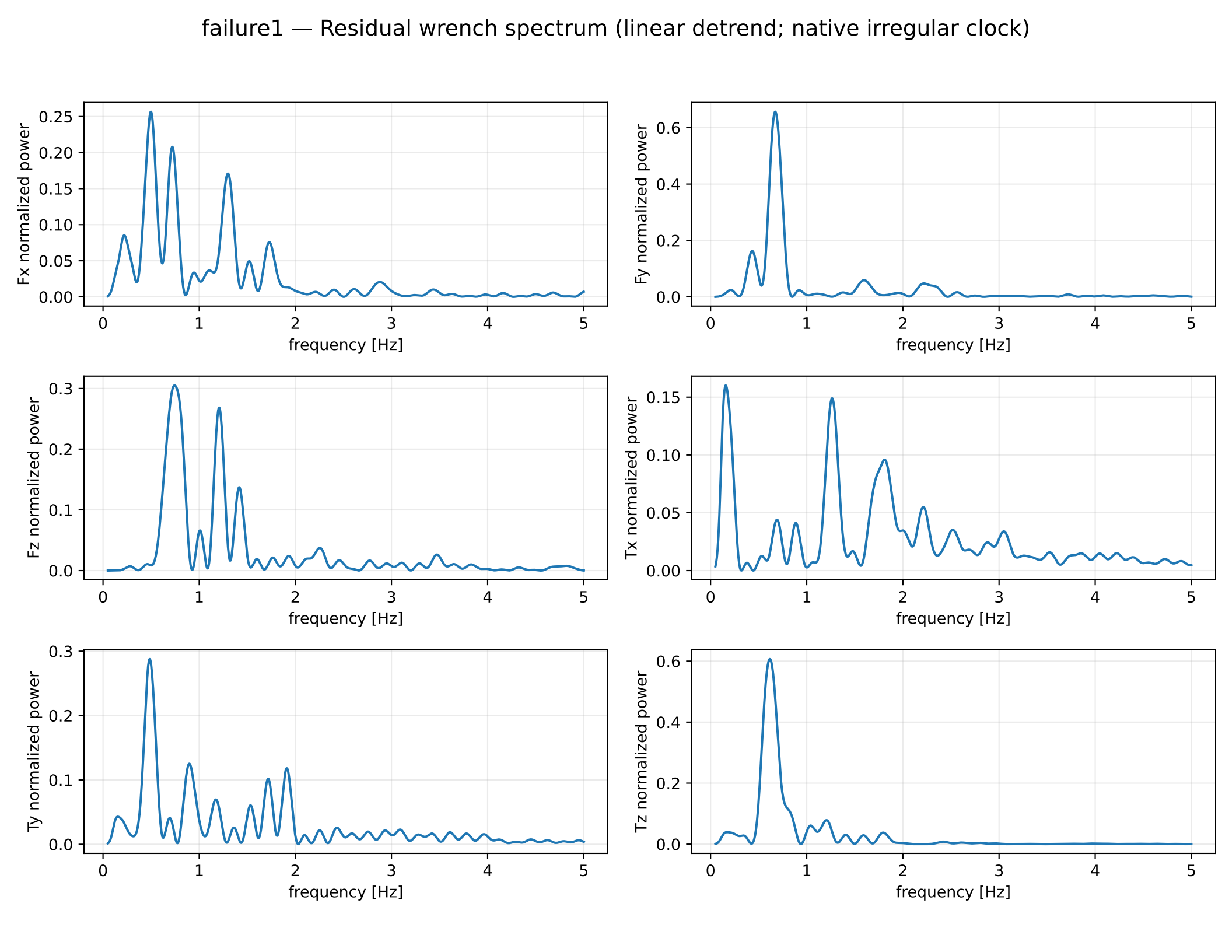
Native IMU support: 19.002181 to 24.997109 bag seconds; N = 1200.

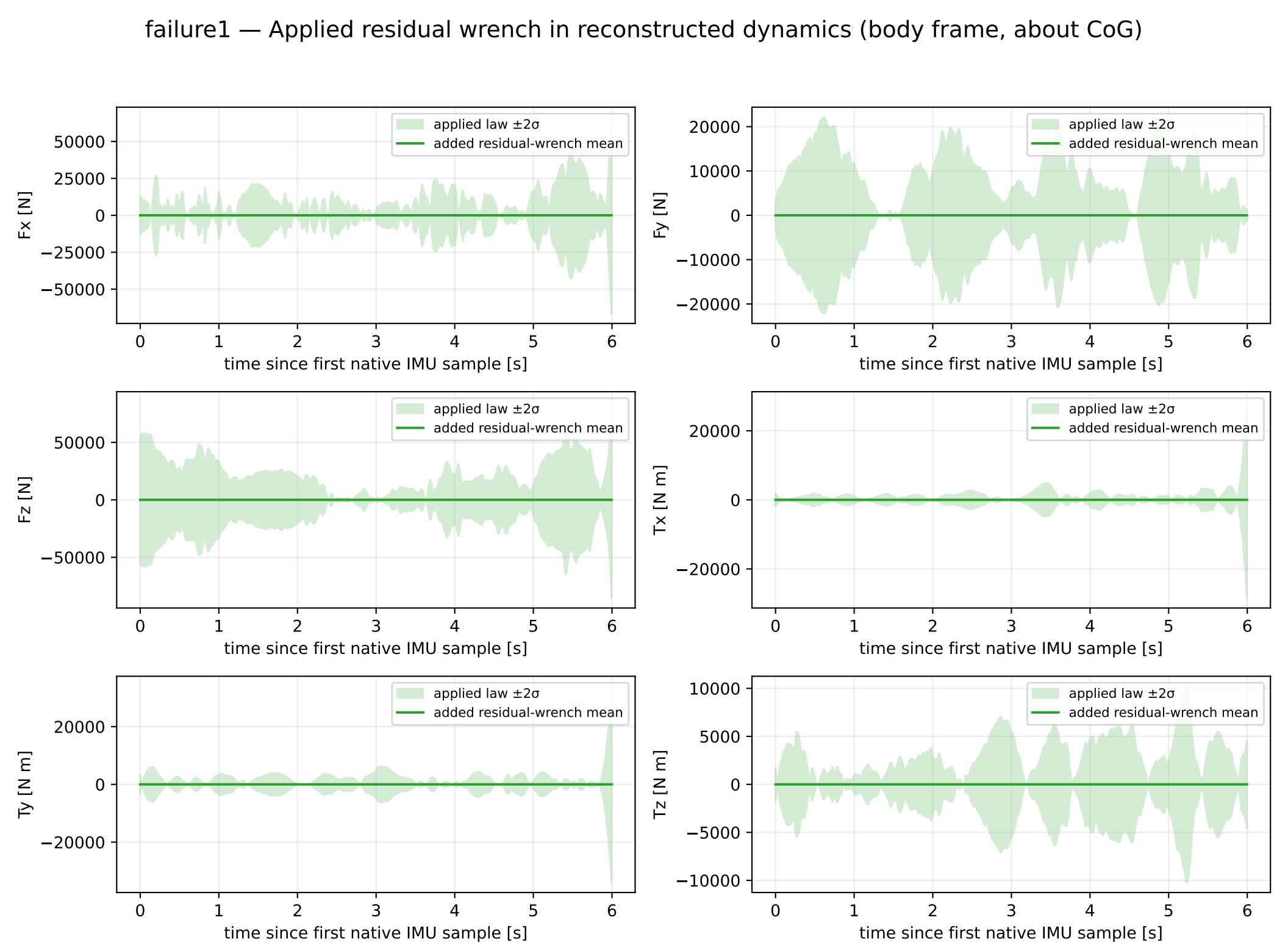
Individual PDF appendices and full-precision covariance CSV contain the same distribution.

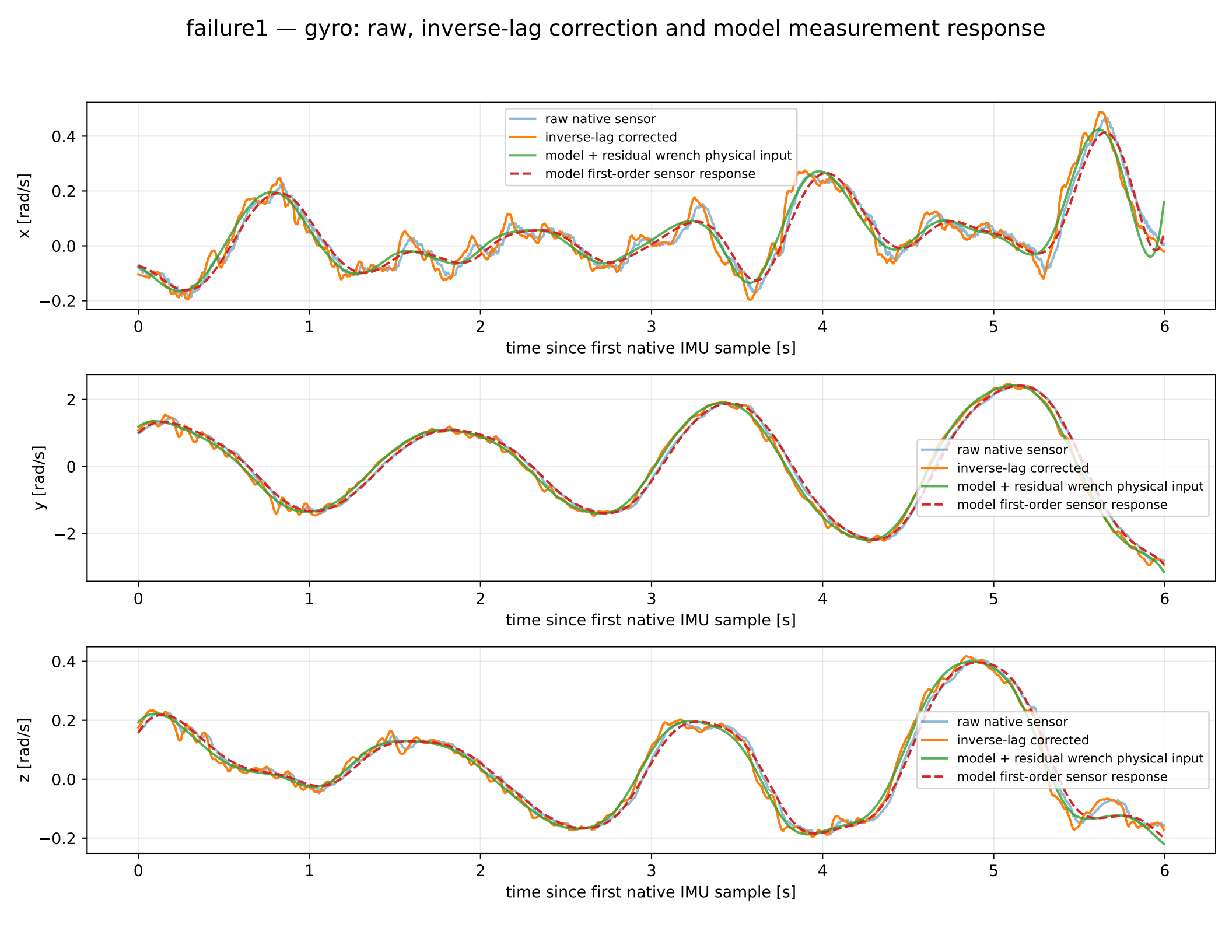

failure1 — Rotor commands and fitted thrust dynamics
rotor 1
rotor 2
rotor 3
rotor 4
sent command
fitted first-order thrust
thrust [N]
time since first native IMU sample [s]
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0

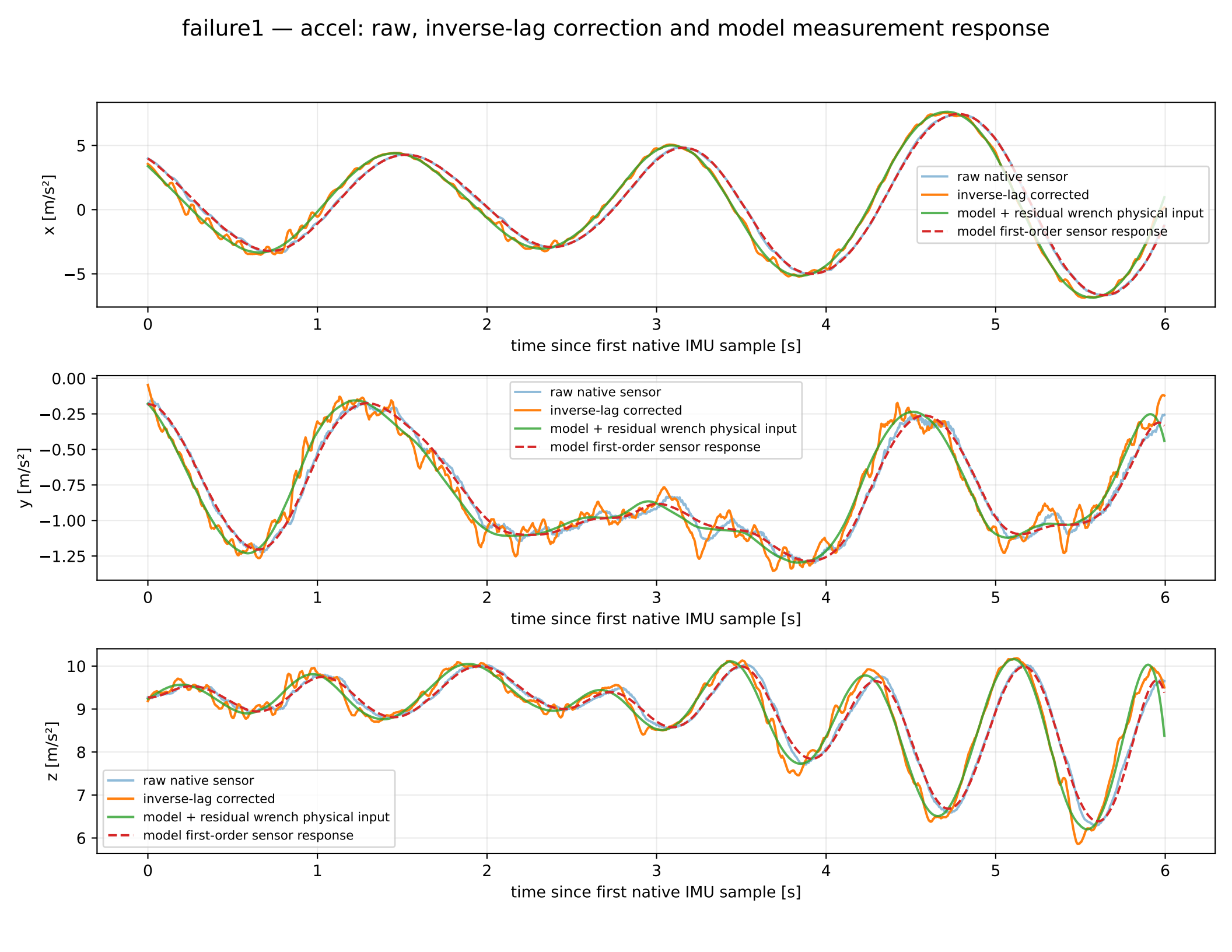

failure1 — Generalized IMU acceleration fit
lag-corrected measurement
identified dynamics
x [m/s²]
x [rad/s²]
y [m/s²]
y [rad/s²]
z [m/s²]
z [rad/s²]
time since first native IMU sample [s]

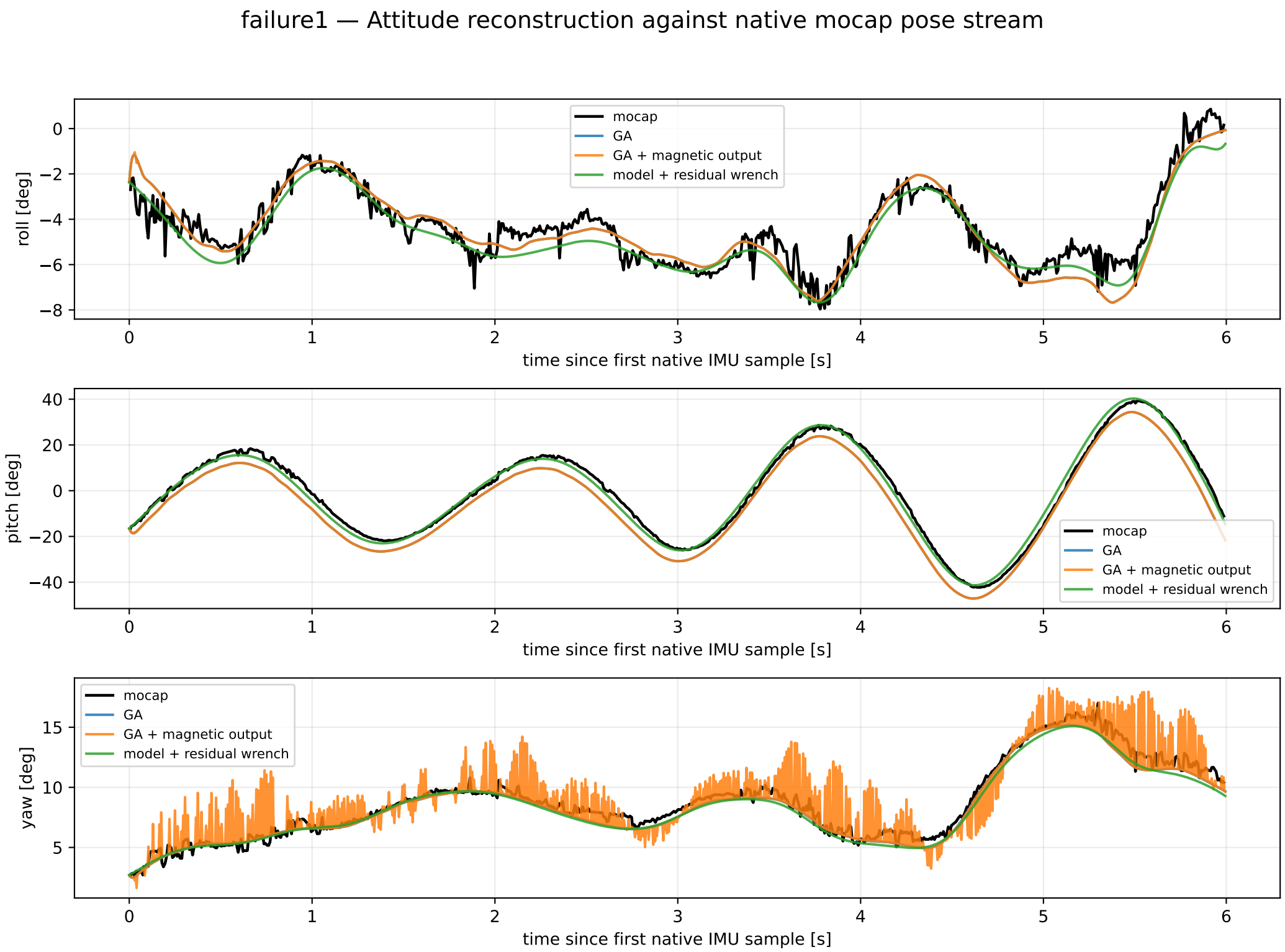

failure1 — Residual wrench at native measurement timestamps
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
time since first native IMU sample [s]

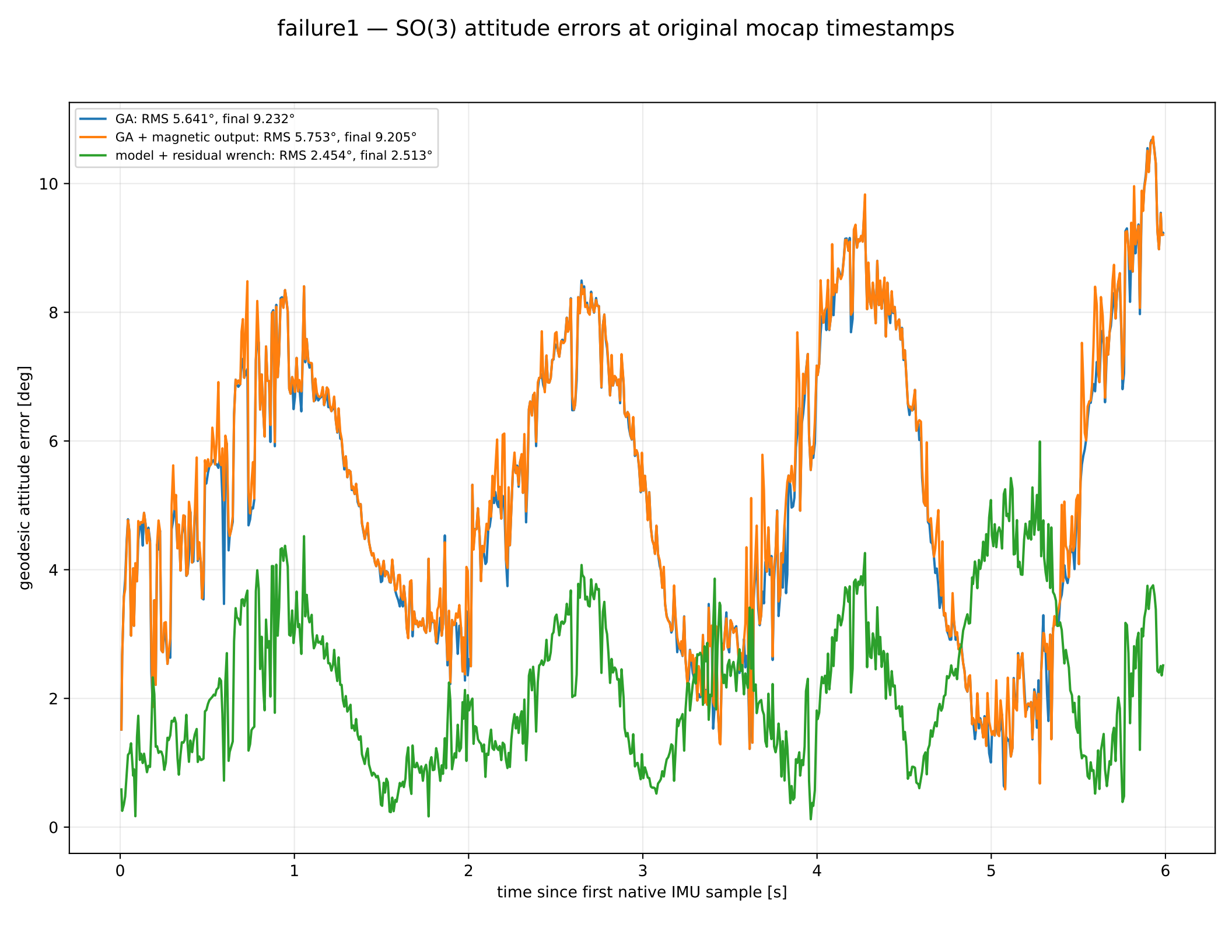

failure1 — Residual wrench spectrum (linear detrend; native irregular clock)
Fx normalized power
Fy normalized power
Fz normalized power
Tx normalized power
Ty normalized power
Tz normalized power
frequency [Hz]

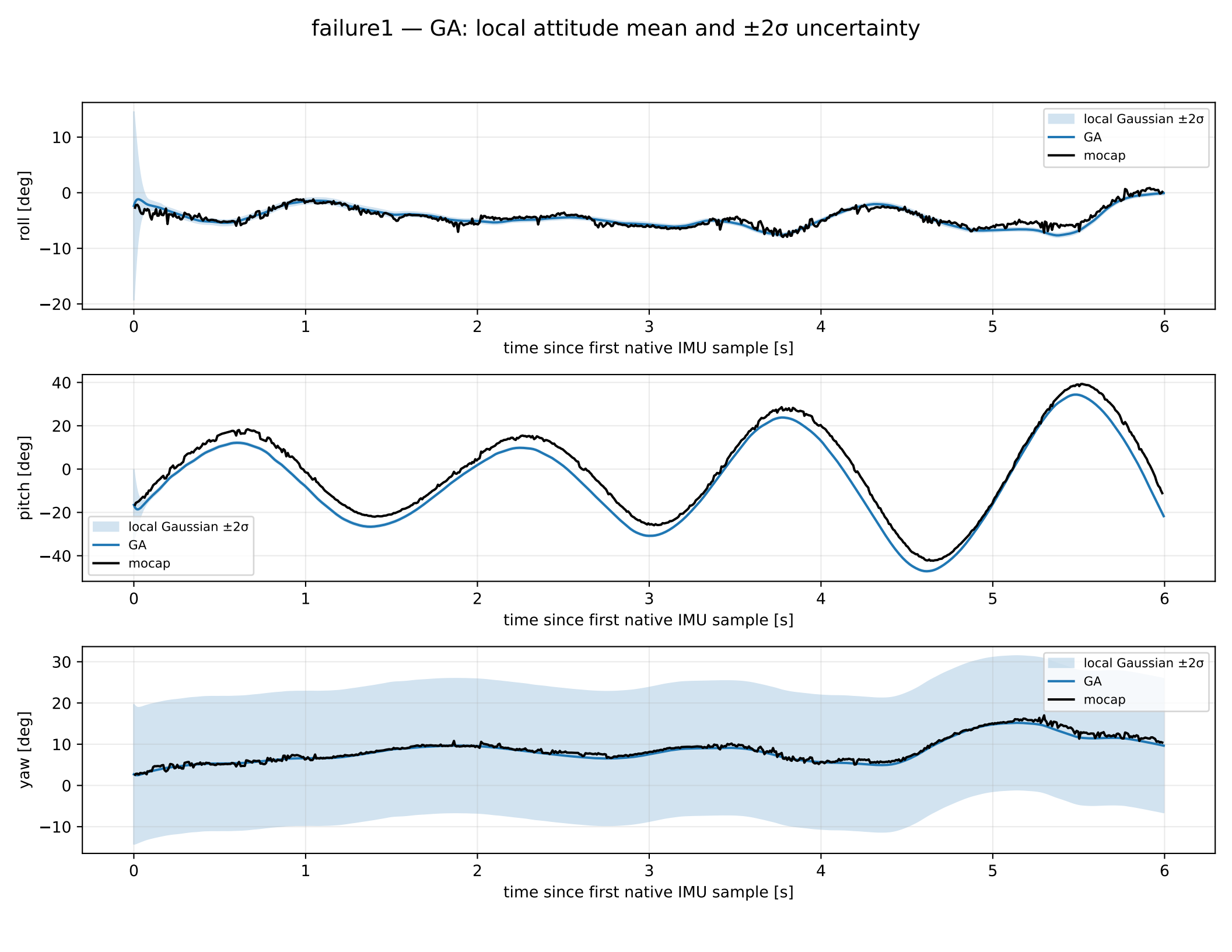

failure1 — Applied residual wrench in reconstructed dynamics (body frame, about CoG)
Fx [N]
Fy [N]
Fz [N]
Tx [N m]
Ty [N m]
Tz [N m]
applied law ±2σ
added residual-wrench mean
time since first native IMU sample [s]
The native-IMU residual wrench is added to the dynamics; this is not a model-only free replay.

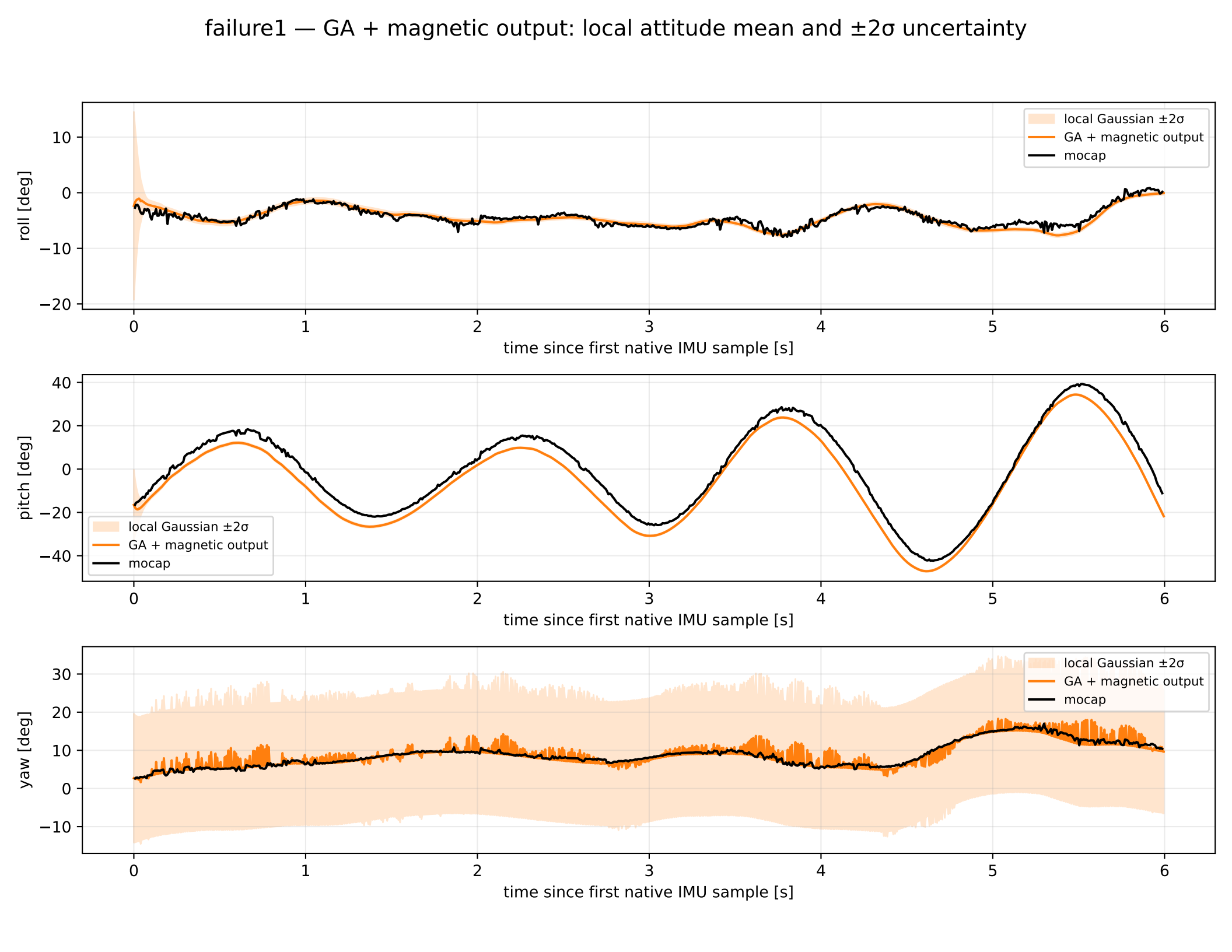

failure1 — gyro: raw, inverse-lag correction and model measurement response
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response
x [rad/s]
y [rad/s]
z [rad/s]
time since first native IMU sample [s]

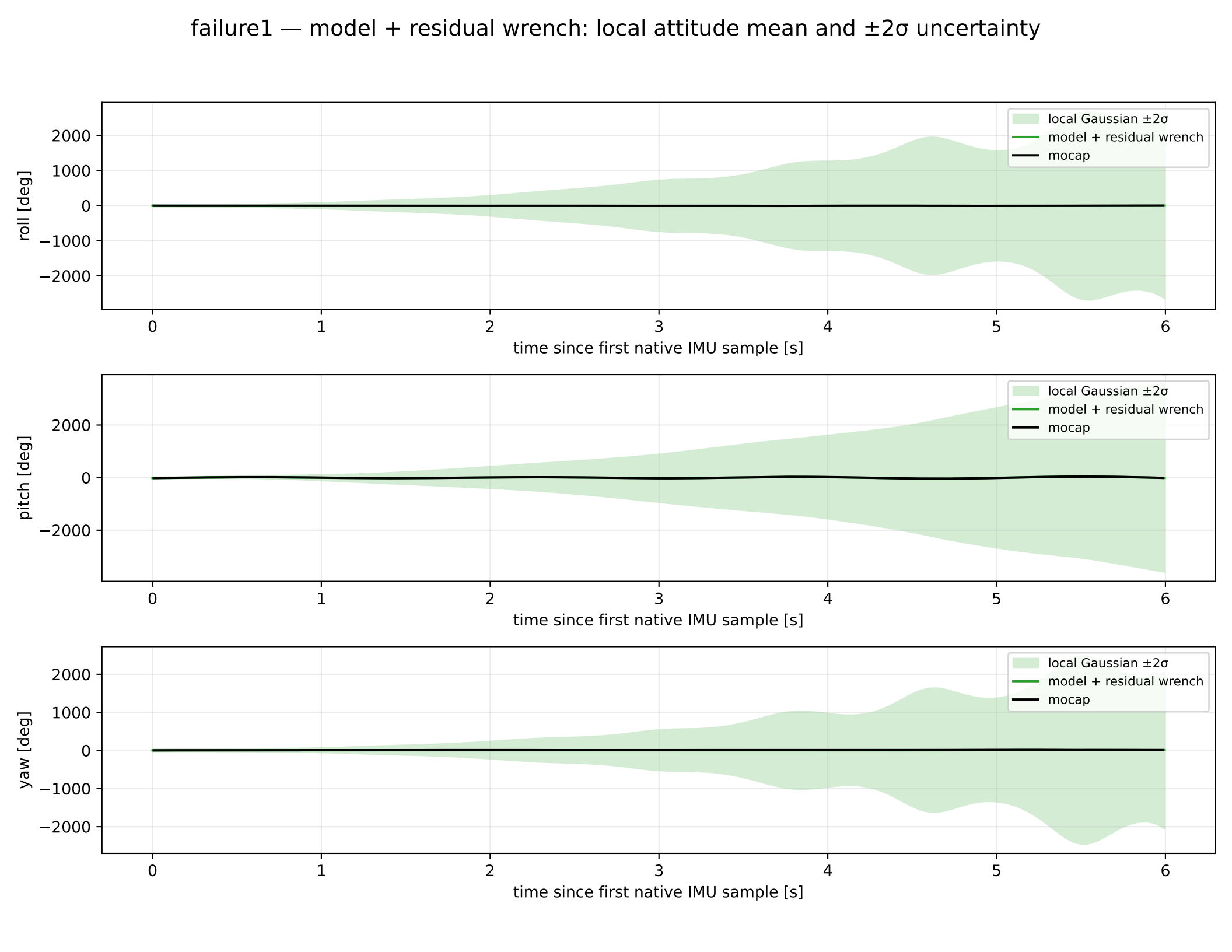

failure1 — accel: raw, inverse-lag correction and model measurement response
x [m/s²]
y [m/s²]
z [m/s²]
time since first native IMU sample [s]
raw native sensor
inverse-lag corrected
model + residual wrench physical input
model first-order sensor response


failure1 — Attitude reconstruction against native mocap pose stream
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
mocap
GA
GA + magnetic output
model + residual wrench

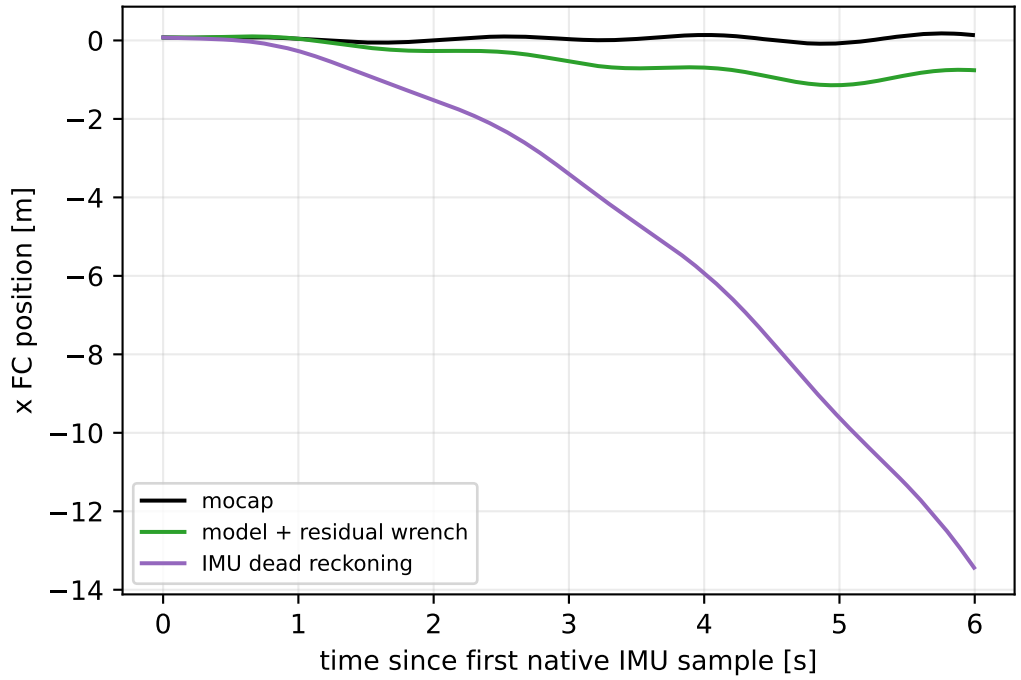

failure1 — SO(3) attitude errors at original mocap timestamps
GA: RMS 5.641°, final 9.232°
GA + magnetic output: RMS 5.753°, final 9.205°
model + residual wrench: RMS 2.454°, final 2.513°
geodesic attitude error [deg]
time since first native IMU sample [s]
0
2
4
6
8
10
0
1
2
3
4
5
6

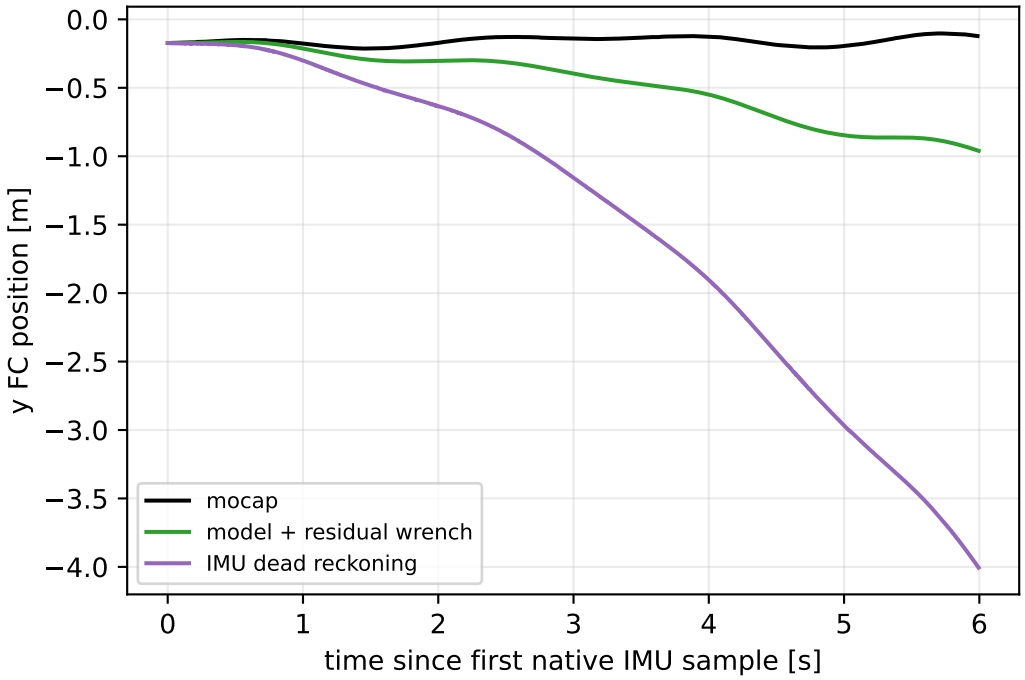

failure1 — GA: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA
mocap

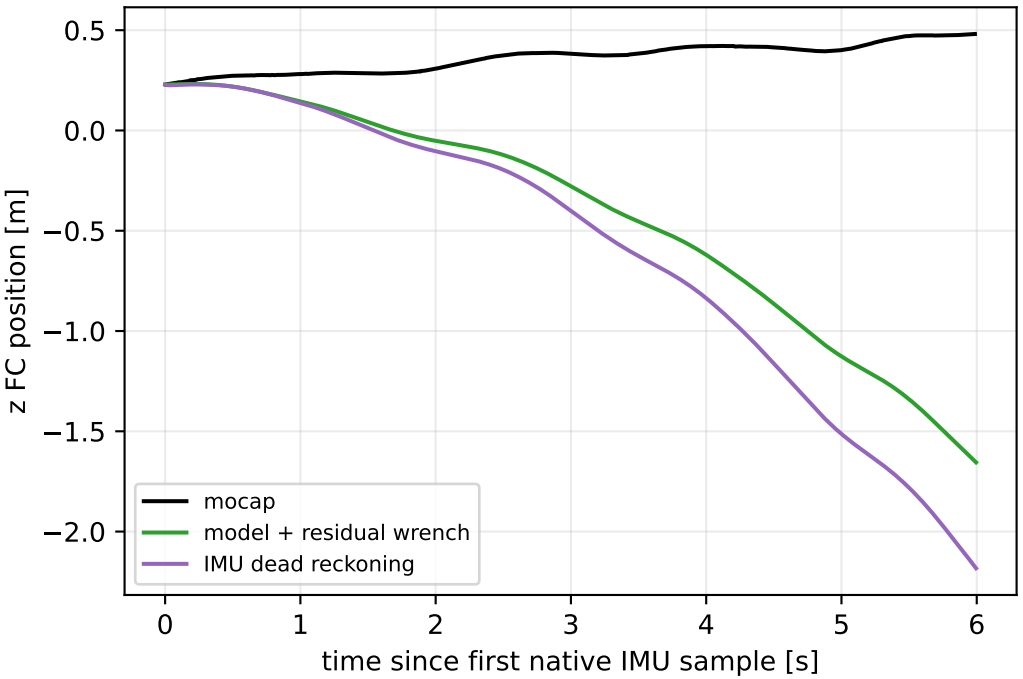

failure1 — GA + magnetic output: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA + magnetic output
mocap

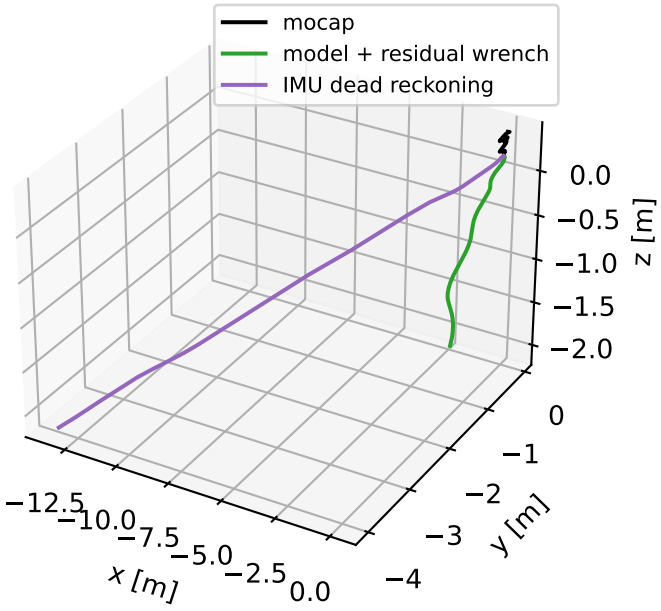

failure1 — model + residual wrench: local attitude mean and ±2σ uncertainty
local Gaussian ±2σ
model + residual wrench
mocap
roll [deg]
pitch [deg]
yaw [deg]
2000
1000
0
−1000
−2000
0
1
2
3
4
5
6
time since first native IMU sample [s]

failure1 — Position reconstruction (mean trajectories)



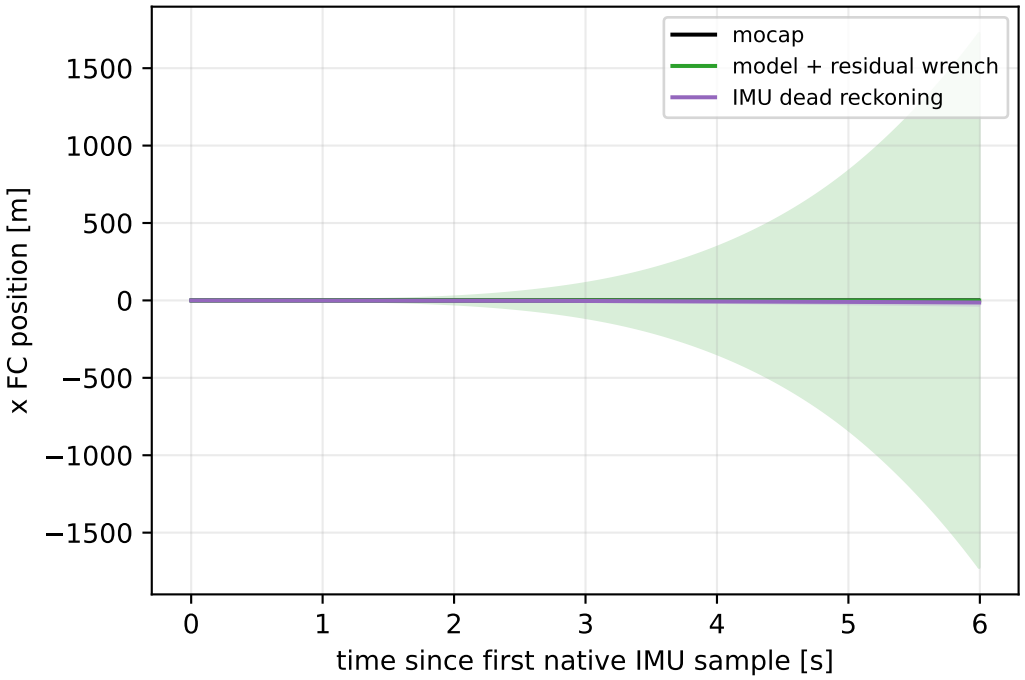


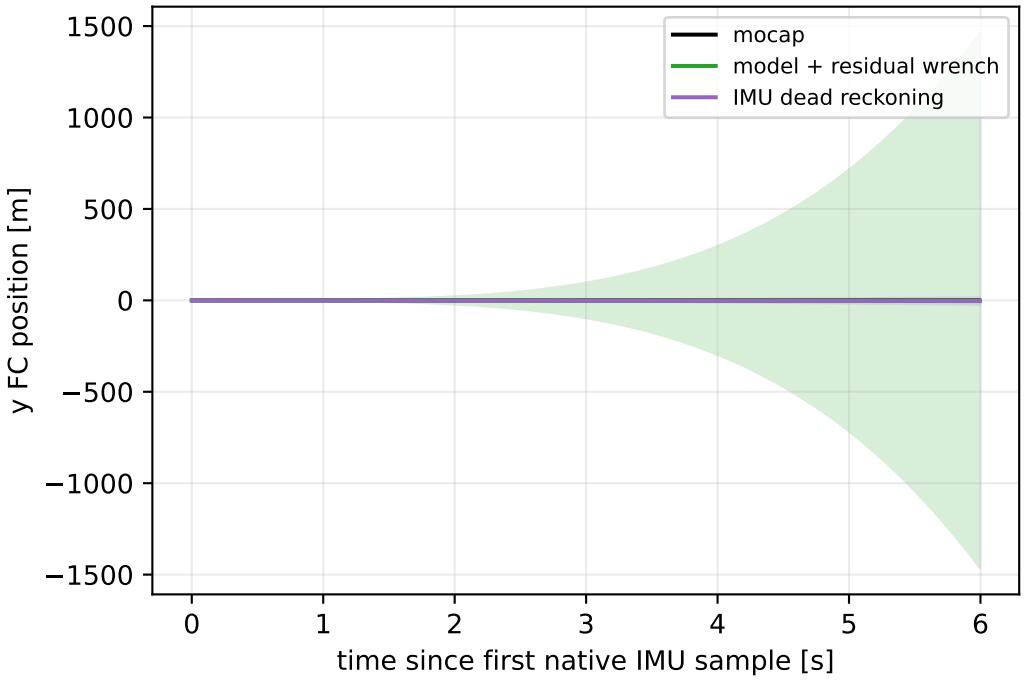


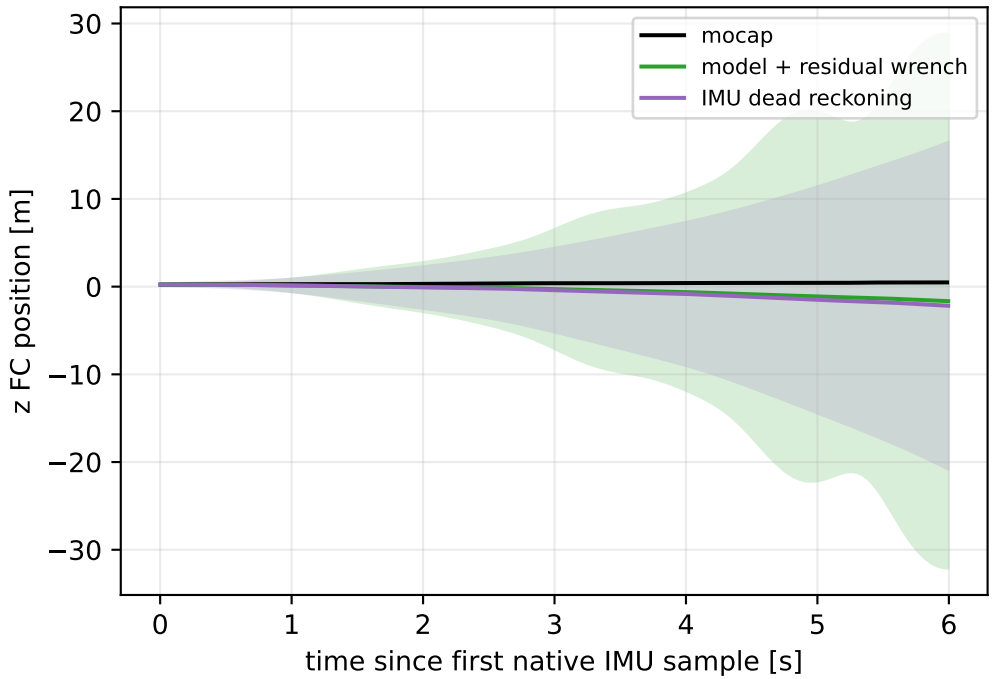


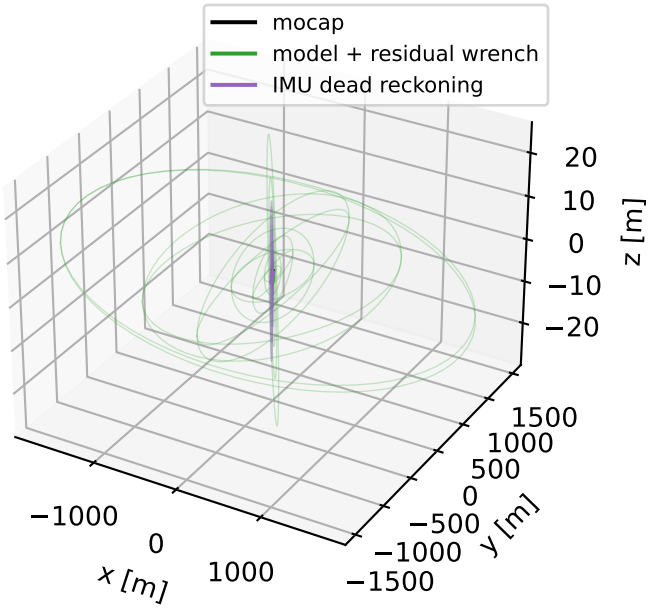
Residual-wrench model replay and IMU integration are not reset to later mocap.

failure1 — Position reconstruction with full propagated ±2σ spread









FC origin; local Gaussian linearization. Ellipsoids: deterministic 2σ factor-SVD wireframes, not samples.

failure1 — Actual estimated SI values and joint covariance coordinate order

#	parameter	unit	estimate	std. deviation
1	mass_kg	kg	5.662356702e+00	1.930790924e+04
2	inertia_xx_kg_m2	kg m^2	3.058171573e-01	1.042792496e+03
3	inertia_yy_kg_m2	kg m^2	3.009270629e-01	1.026123399e+03
4	inertia_zz_kg_m2	kg m^2	6.044695457e-01	2.061189733e+03
5	inertia_xy_kg_m2	kg m^2	3.665243642e-03	1.249786514e+01
6	inertia_xz_kg_m2	kg m^2	-1.505029697e-03	5.134717106e+00
7	inertia_yz_kg_m2	kg m^2	1.856582079e-02	6.330401638e+01
8	cog_x_m	m	2.481261142e-02	1.549734871e-03
9	cog_y_m	m	-5.845013145e-03	9.014727386e-04
10	cog_z_m	m	-2.434450651e-02	1.074343382e-03
11	force_effectiveness_1	dimensionless	1.814096850e+00	6.185834614e+03
12	force_effectiveness_2	dimensionless	2.485234671e+00	8.474329527e+03
13	force_effectiveness_3	dimensionless	2.172043873e+00	7.406390656e+03
14	force_effectiveness_4	dimensionless	1.720953263e+00	5.868230662e+03
15	thrust_time_constant_seconds	s	2.190042293e-01	4.009869312e-03
16	gyro_time_constant_seconds	s	3.719012789e-02	1.892996386e-03
17	accel_time_constant_seconds	s	6.620588407e-02	3.042504690e-04

Common-scale gauge: physical chart projected onto existing 13-D orthogonal common-scale section; tau coordinates unaffected

Covariance units are products of row/column units. All cross terms and weak modes are retained.

Full-precision values: parameter_values_si.csv; full 17×17 covariance: parameter_covariance_si.csv.

failure1 — Full SI covariance: rows 1–9, columns 1–9

	1	2	3	4	5	6	7	8	9
1	3.727954e+08	2.013414e+07	1.981230e+07	3.979726e+07	2.413076e+05	-9.914038e+04	1.222268e+06	3.445390e-01	3.664392e+00
2	2.013414e+07	1.087416e+06	1.070034e+06	2.149393e+06	1.303268e+04	-5.354430e+03	6.601295e+04	1.861156e-02	1.979035e-01
3	1.981230e+07	1.070034e+06	1.052929e+06	2.115035e+06	1.282435e+04	-5.268839e+03	6.495773e+04	1.831100e-02	1.947444e-01
4	3.979726e+07	2.149393e+06	2.115035e+06	4.248503e+06	2.576047e+04	-1.058360e+04	1.304816e+05	3.678449e-02	3.911820e-01
5	2.413076e+05	1.303268e+04	1.282435e+04	2.576047e+04	1.561966e+02	-6.417284e+01	7.911651e+02	2.224373e-04	2.372069e-03
6	-9.914038e+04	-5.354430e+03	-5.268839e+03	-1.058360e+04	-6.417284e+01	2.636532e+01	-3.250473e+02	-8.757603e-05	-9.765488e-04
7	1.222268e+06	6.601295e+04	6.495773e+04	1.304816e+05	7.911651e+02	-3.250473e+02	4.007398e+03	1.130850e-03	1.201249e-02
8	3.445390e-01	1.861156e-02	1.831100e-02	3.678449e-02	2.224373e-04	-8.757603e-05	1.130850e-03	2.401678e-06	-4.419781e-07
9	3.664392e+00	1.979035e-01	1.947444e-01	3.911820e-01	2.372069e-03	-9.765488e-04	1.201249e-02	-4.419781e-07	8.126531e-07

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure1 — Full SI covariance: rows 1–9, columns 10–17

	10	11	12	13	14	15	16	17
1	-3.212669e+00	1.194355e+08	1.636216e+08	1.430019e+08	1.133033e+08	-7.492179e+00	6.046955e+00	1.781598e-01
2	-1.735176e-01	6.450542e+06	8.836967e+06	7.723329e+06	6.119347e+06	-4.046489e-01	3.265867e-01	9.622524e-03
3	-1.707430e-01	6.347430e+06	8.695708e+06	7.599871e+06	6.021529e+06	-3.981794e-01	3.213651e-01	9.468692e-03
4	-3.429751e-01	1.275018e+07	1.746720e+07	1.526598e+07	1.209554e+07	-7.998307e-01	6.455322e-01	1.901997e-02
5	-2.079012e-03	7.730973e+04	1.059110e+05	9.256407e+04	7.334036e+04	-4.849407e-03	3.914340e-03	1.152546e-04
6	8.540140e-04	-3.176243e+04	-4.351317e+04	-3.802962e+04	-3.013162e+04	2.002309e-03	-1.609694e-03	-4.723286e-05
7	-1.053325e-02	3.915882e+05	5.364591e+05	4.688543e+05	3.714826e+05	-2.456180e-02	1.982510e-02	5.839530e-04
8	-1.367391e-07	1.103637e-01	1.512479e-01	1.321798e-01	1.046921e-01	-1.749291e-06	2.582529e-07	9.958268e-08
9	1.834891e-07	1.173989e+00	1.608306e+00	1.405639e+00	1.113724e+00	-6.724622e-07	5.948103e-07	4.787001e-08

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure1 — Full SI covariance: rows 10–17, columns 1–9

	1	2	3	4	5	6	7	8	9
10	-3.212669e+00	-1.735176e-01	-1.707430e-01	-3.429751e-01	-2.079012e-03	8.540140e-04	-1.053325e-02	-1.367391e-07	1.834891e-07
11	1.194355e+08	6.450542e+06	6.347430e+06	1.275018e+07	7.730973e+04	-3.176243e+04	3.915882e+05	1.103637e-01	1.173989e+00
12	1.636216e+08	8.836967e+06	8.695708e+06	1.746720e+07	1.059110e+05	-4.351317e+04	5.364591e+05	1.512479e-01	1.608306e+00
13	1.430019e+08	7.723329e+06	7.599871e+06	1.526598e+07	9.256407e+04	-3.802962e+04	4.688543e+05	1.321798e-01	1.405639e+00
14	1.133033e+08	6.119347e+06	6.021529e+06	1.209554e+07	7.334036e+04	-3.013162e+04	3.714826e+05	1.046921e-01	1.113724e+00
15	-7.492179e+00	-4.046489e-01	-3.981794e-01	-7.998307e-01	-4.849407e-03	2.002309e-03	-2.456180e-02	-1.749291e-06	-6.724622e-07
16	6.046955e+00	3.265867e-01	3.213651e-01	6.455322e-01	3.914340e-03	-1.609694e-03	1.982510e-02	2.582529e-07	5.948103e-07
17	1.781598e-01	9.622524e-03	9.468692e-03	1.901997e-02	1.152546e-04	-4.723286e-05	5.839530e-04	9.958268e-08	4.787001e-08

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.

failure1 — Full SI covariance: rows 10–17, columns 10–17

	10	11	12	13	14	15	16	17
10	1.154214e-06	-1.029271e+00	-1.410060e+00	-1.232360e+00	-9.764187e-01	4.233956e-07	6.557444e-07	-3.506045e-08
11	-1.029271e+00	3.826455e+07	5.242080e+07	4.581471e+07	3.629990e+07	-2.400313e+00	1.937305e+00	5.707725e-02
12	-1.410060e+00	5.242080e+07	7.181426e+07	6.276420e+07	4.972932e+07	-3.288367e+00	2.654033e+00	7.819603e-02
13	-1.232360e+00	4.581471e+07	6.276420e+07	5.485462e+07	4.346241e+07	-2.873972e+00	2.319578e+00	6.834223e-02
14	-9.764187e-01	3.629990e+07	4.972932e+07	4.346241e+07	3.443613e+07	-2.277082e+00	1.837848e+00	5.414755e-02
15	4.233956e-07	-2.400313e+00	-3.288367e+00	-2.873972e+00	-2.277082e+00	1.607905e-05	-5.686609e-06	-7.164284e-07
16	6.557444e-07	1.937305e+00	2.654033e+00	2.319578e+00	1.837848e+00	-5.686609e-06	3.583435e-06	2.708871e-07
17	-3.506045e-08	5.707725e-02	7.819603e-02	6.834223e-02	5.414755e-02	-7.164284e-07	2.708871e-07	9.256835e-08

Indices refer to the preceding SI parameter table; Cij has units unit(i) × unit(j).

Actual joint covariance, including all sensor-lag cross covariance; no eigenvalue clipping.